

COUNCIL MEMBERS

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RECEIVED
 JUL 05 1995

SOLANO COUNTY
 Emergency Medical Services

CITY OF VACAVILLE

650 MERCHANT STREET, VACAVILLE, CALIFORNIA 95688

ESTABLISHED 1850

June 30, 1995

Ron Lopez, Pre-Hospital Care Coordinator
 Solano County EMSA
 717 Missouri Street, Suite B
 Fairfield, CA 94533

Dear Mr. Lopez:

This letter is a request that County EMSA take a serious look at the inconsistencies that occurs in the treatment, transportation, and destinations of both trauma and medical calls for service. Over the past ten years, the County has been plagued with inconsistent, opposing, nonsensical protocols and memorandums. These discrepancies are not the fault of one person, organization or company, but a product of lack of unity among all players in pre-hospital care; hospital staff, physician groups, and administration of the County's Emergency Medical System. How can one county have different trauma criteria, trauma destinations, and treatment protocols?

Recently, one such misunderstanding took place between NorthBay Medical Center and the Vacaville Fire Department. Rescue Four responded to a 43 year old male who was complaining of an altered level of consciousness. He is a known diabetic. See attached copy of PCR #53555. Basically, the patient had a blood sugar level of 43. Oral glucose was given and the blood sugar rose to 54. At that time, base contact was made and the paramedic requested an IV of Normal Saline, and one-half amp of 50% Dextrose IVP. This would raise the patient's blood sugar level over 80. The MICN at the base hospital stated that once ALS procedures have been started the patient had to be transported to a hospital. The paramedic disagreed with this decision and asked to talk directly to the physician. Once the physician got on the phone, he reiterated what the MICN said and ordered the paramedic to transport the patient to the hospital. The paramedic explained to the physician that it was not possible to transport the patient if he stated he did not want to go. The physician then talked directly to the patient and the patient finally reluctantly agreed to be transported. At the receiving hospital, an emergency room nurse asked why this patient was transported to the hospital with a blood sugar now of 104. The paramedic attempted to explain. The patient was given some orange juice and released less than one hour later.

DEPARTMENTS: Area Code (707)

Building 449-5152	City Attorney 449-5105	City Manager 449-5100	Finance 449-5117	Fire 449-1838	Personnel 449-5101	Planning 449-5140	Police 449-5200	Public Works 449-5170	Community Services 449-5654
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The point missed was that the patient was not able to take care of himself until the oral glucose, and for that matter, IV Dextrose, raised his blood sugar to a level that was consistent with rational behavior and full orientation to his surroundings. If a patient refuses service because he is aware of his/her chronic medical condition, the EMS system has no authority to order him/her to be transported to a hospital.

We constantly talk about unnecessary IV's, unwarranted helicopter transports, and keeping the medical costs down for the people we treat or transport. How can we talk one way and act another?

I am requesting that all interested or connected parties sit down and discuss a way to provide simple, consistent, and safe treatment, as well as the transportation of those patients who request our medical expertise in Solano County.

Please contact me if you have any questions or comments. Thank you for your time and commitment.

Sincerely,



Rob Glankler, EMS Coordinator
Vacaville Fire Department

Enclosure (PCR #53555)

ems-let.doc

Let 7.10.95



Solano County Health and Social Services Department

Mental Health Services
Public Health Services

Adult and Child Services
Substance Abuse Services

Eligibility and Employment Services
Public Guardian/Conservator

Donald R. Rowe, Director

6 July, 1995

Mr. Rob Glankler, EMS Coordinator
Vacaville Fire Department
650 Merchant St.
Vacaville, Ca. 95688

Dear Mr. Glankler:

Thank you for letter of June 30 expressing your concern over the inconsistencies in policies, procedures, and management strategies for patients our EMS system serves. I know from my own personal field experiences (long ago and far away that they are!) that this is a problem inherent to most, and probably all, EMS Systems. It is my feeling that if we look closely at these and similar problems we will find that our frustration stems from the difference between our expectations for patient management and reality, and from the difference between our expectation of what an EMS System should look like and what we actually see. I am reminded of the time I received an *educational lecture* from an MICN about why we do not administer prophylactic lidocaine in the presence of chest pain, when the patient, on the next bed, suddenly (and without asking permission, I might add!) went into V-Tach and subsequently arrested.

I think its safe to say that we're all familiar with the problem. The real question is "What are we going to do about it?" Our typical response, and one you mentioned in your letter, is to produce "...inconsistent, opposing, nonsensical protocols and memorandums." Fortunately, you also mention the solution, forinterested or connected parties to sit down and discuss...That would be the essence of a system wide Quality Improvement (QI) committee.

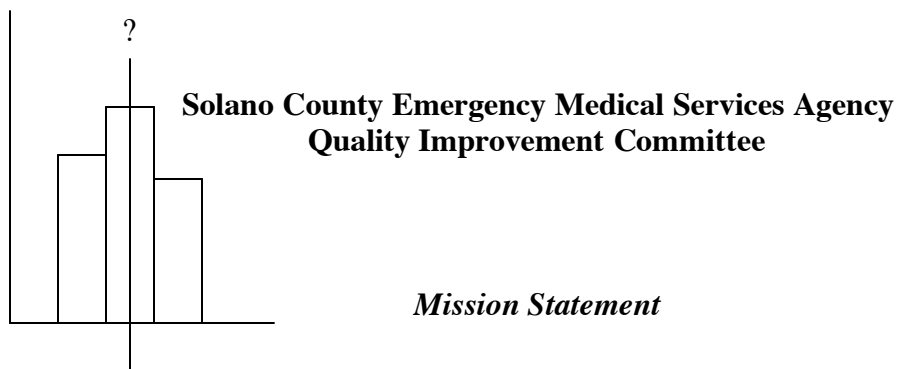
Similar interest has also been expressed by personnel from both Medic and Baystar, and I would like to invite you to attend an ad hoc committee within the next several weeks (To Be Announced) where we can begin to develop a legitimate process to implement QI in Solano County.

I thank you for your comments and assure you that others, myself included, share in your frustration. I look forward to working with you and others on this worthwhile endeavor.

Sincerely,

A handwritten signature in cursive script, appearing to read "Michael A. Frenn".

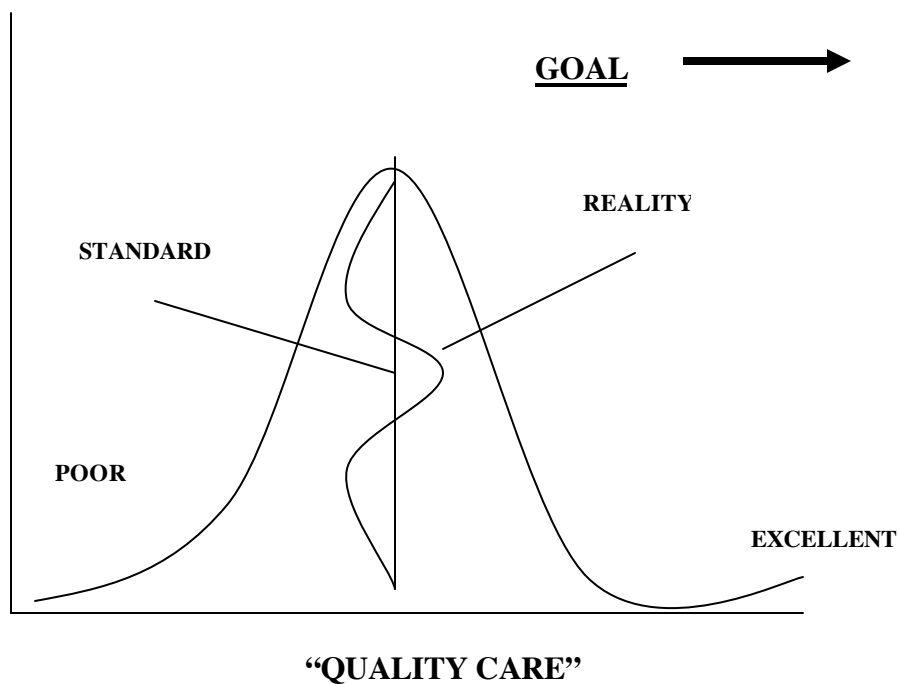
Michael A. Frenn, EMS Administrator
Solano County EMS Agency

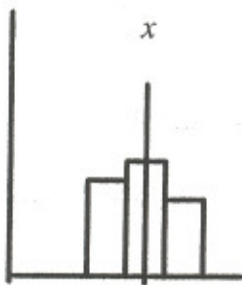


The mission of the Quality Improvement committee is to help define “effectiveness” of the prehospital care system in Solano County by developing, through a peer-based process, consensus on community standards of care, evaluating how closely the system meets these standards, and implementing action plans designed to result in system performance that consistently meets or exceeds these standards.

Quality Improvement Project

General Principles





Solano County Quality Improvement Committee

PARAMETER: Patient Assessment

STANDARD: All persons with a health complaint shall receive a physical assessment that at a minimum evaluates the following:

- Chief Complaint
- Characteristics and onset of condition
- Medications
- Vital signs (P, BP, R)
- Primary Assessment
- Secondary Assessment

Assessment priorities should be consistent with the attached algorithm.

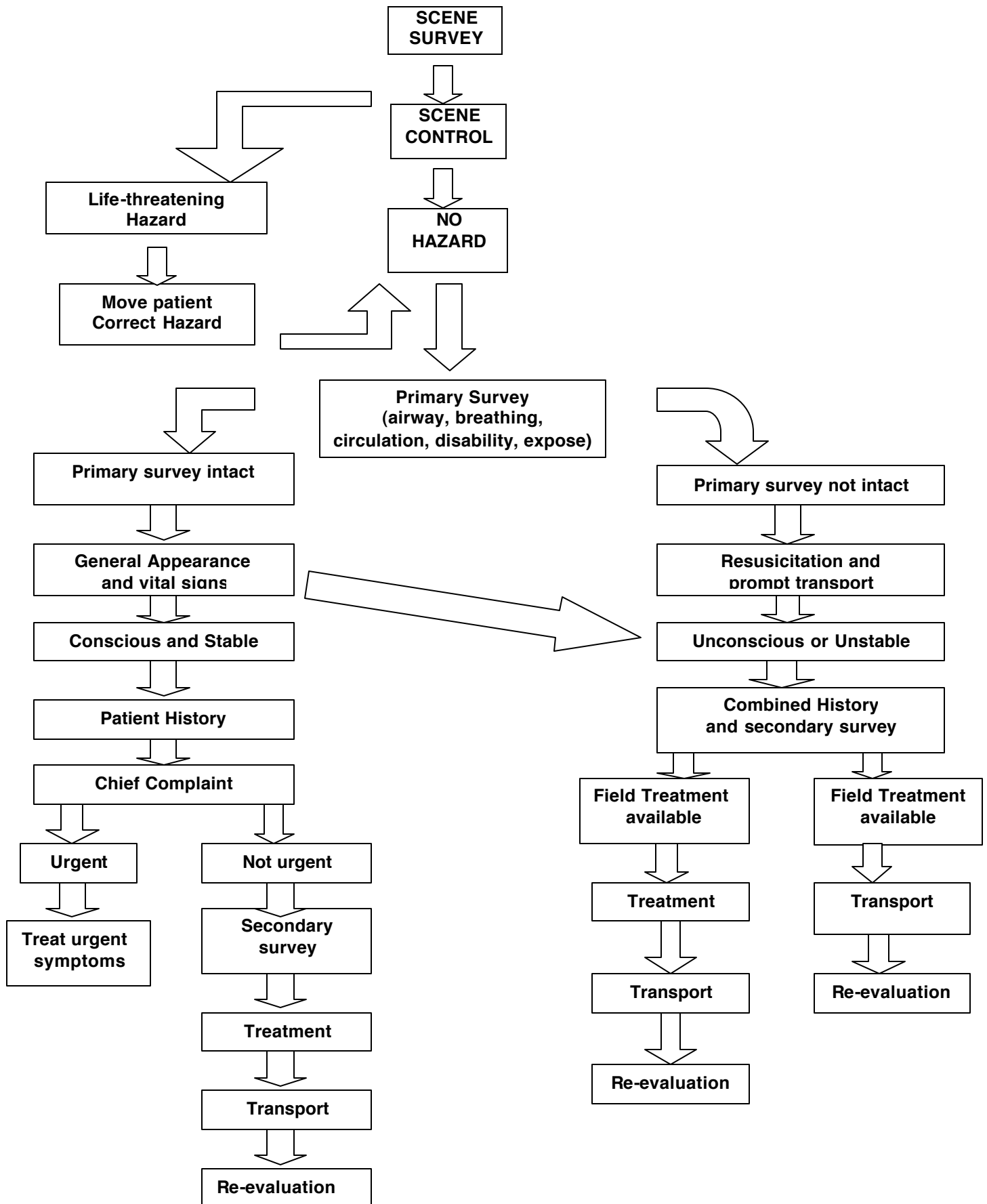
REFERENCE: Manual of Advanced Prehospital Care, 2nd. ed. (1984)
Mosby's Paramedic Textbook (1994)

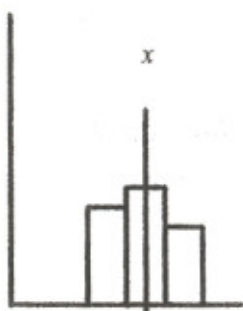
Dr. Thomas L. Charron, Health Officer/EMS Agency Medical Director

9-21-00
Effective Date

Dr. Allen Morini, Assistant EMS Agency Medical Director

9-21-00
Distribution Date





Solano County Quality Improvement Committee

PARAMETER: General Patient Assessment

STANDARD: All persons with a health complaint shall receive a physical assessment that at a minimum evaluates the following:

- Primary Assessment
 - Airway
 - Breathing
 - Circulation
 - LOC (Level of Consciousness)
 - Chief Complaint
- Secondary Assessment
 - Vital Signs (Pulse, BP, Respiratory Rate, Skin Signs)
 - History of Present Illness
 - Past Medical History
 - Medications
 - Allergies
 - Complete Problem-Oriented Physical Exam
 - Reassessment of Vital Signs

REFERENCE: Manual of Advanced Prehospital Care, 2nd. ed. (1984)

Thomas L. Charron MD

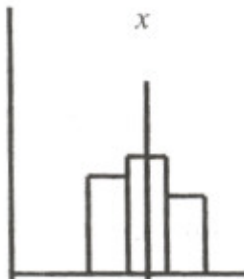
Thomas L. Charron, MD, MPH, Health Officer,
EMS Agency Medical Director

10/30/97
Effective Date

Allen Morini, DO

Allen Morini, DO, Assistant EMS Agency Medical Director

10/30/97
Distribution Date



Solano County Quality Improvement Committee

PARAMETER: Pre Hospital use of Automatic External Defibrillation (AED)

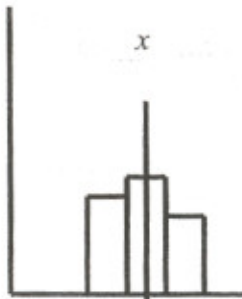
STANDARD: All First Responder and EMT-1 personnel trained and authorized to treat patients using automatic defibrillation equipment shall adhere to the following standards:

- The EMT-D medical record was legible and complete.
- Treatment and/or care provided by the EMT-D was appropriately documented.
- Patient response to EMT-D treatment was appropriately documented.
- Defibrillator electronic analysis began within 90 seconds of arrival at patient.
- Time of collapse was recorded.
- Placement of Defibrillator pads was documented.
- Defibrillation was performed consistent with Solano EMS policy.
- Documentation of vital signs.
- Documentation of pulses checked during CPR chest compressions.
- EMT-D rescue crew responded within 2 minutes of receiving the call.
- Response time was within 5 minutes.
- Documentation of aid before arrival of EMT-D unit.
- Treatment outcome documented.

REFERENCE: SEMSC Agency Policy 3300 EMT-1 Defibrillation Authorization.
SEMSC Agency Policy 3301 Public Safety / First Responder
Defibrillation Authorization.

Dr. Thomas L. Charron, Health Officer/EMS Agency Medical Director
Effective Date 9-21-00

Dr. Allen Morini, Assistant EMS Agency Medical Director
Distribution Date 9-21-00



Solano County Quality Improvement Committee

PARAMETER: Chest Pain

STANDARD: All patients who present with the complaint of chest pain will have the following minimum documentation:

- Description of pain to include:
 - Provocation / Precipitating factors.
 - Quality.
 - Radiation.
 - Severity.
 - Time of pain.
- Past medical history.
- Cardiac risk factors.
- History of drug use related to chief complaint.
- ECG rhythm.
- Associated Shortness of Breath.
- Medications noted.
- Allergies noted.
- Selected and followed appropriate protocol.

REFERENCE: Bryan E. Bledsoe, Robert S. Porter, Bruce R. Shade, *Paramedic Emergency Care* 1991 Prentice-Hall.

Thomas Charron MD

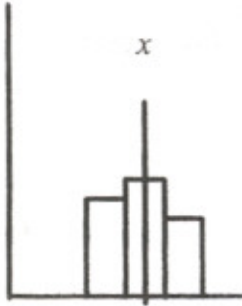
Dr. Thomas L. Charron, Health Officer/EMS Agency Medical Director

Effective Date 09-21-00

A. Morini, DO

Dr. Allen Morini, Assistant EMS Agency Medical Director

Distribution Date 09-21-00



Solano County Quality Improvement Committee

PARAMETER: General Hospital Notification Report

STANDARD: The format for making this type of base contact should include, at a minimum, the following information:

- Name of person making contact
- Solano County identifier of field unit
- Purpose of the contact
- Transport destination
- Age, Sex, Chief Complaint of the patient
- Clinical Impression
- ETA

REFERENCE: Solano County CQI Committee, 9/18/97.

Thomas Charron MD

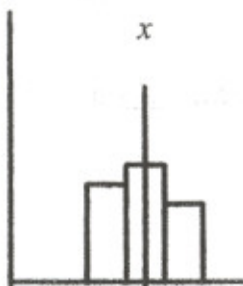
Thomas L. Charron, MD, MPH, Health Officer,
EMS Agency Medical Director

10/30/97
Effective Date

Allen Morini, DO

Allen Morini, DO, Assistant EMS Agency Medical Director

10/30/97
Distribution Date



Solano County Quality Improvement Committee

PARAMETER: Base Contact for Medical Consultation

STANDARD: Contact with the base for medical consultation should include, at a minimum, the following information:

- Name of person making the contact
- Solano County identifier of field unit
- Whether MICN or Physician contact is desired
- Nature of the consultation (AMA, DNR, Treatment Orders, etc.)
- Transport destination
- Age, Sex, Chief Complaint of patient
- Clinical Impression
- Treatment rendered (BLS and ALS)
- Patient's response to treatment
- Requests for medical orders must also include:
 - ☛ Level of Consciousness (LOC)
 - ☛ Vital signs
 - ☛ Pertinent Medical History
 - ☛ Medications
 - ☛ Allergies
 - ☛ Pertinent Physical Exam findings (level of distress, lung sounds, skin signs)
- ETA

REFERENCE: Solano County CQI Committee, 9/18/97

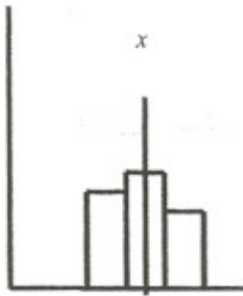
Thomas L. Charron MD

Thomas L. Charron, MD, MPH, Health Officer,
EMS Agency Medical Director

10/30/97
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Allen Morini, DO
Allen Morini, DO, Assistant EMS Agency Medical Director

10/30/97
Distribution Date



Solano County Quality Improvement Committee

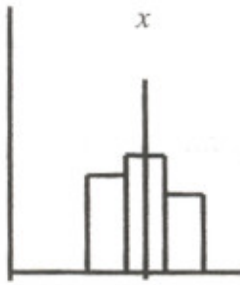
PARAMETER: Documentation of the PCR. Subtopic: PCR legibility, spelling.

STANDARD: The following are minimum documentation standards when completing Patient Care Reports:

- Printed and/or blocked lettering.
- Accurate, complete documentation in all fields to include the MICN and EMS personnel participating in the call.
- Accuracy in spelling.

Dr. Thomas L. Charron, Health Officer/EMS Agency Medical Director
Effective Date 09-21-00

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Solano County Quality Improvement Committee

PARAMETER: PCR Documentation.
Subtopic: Documenting the use of Mechanical Restraints.

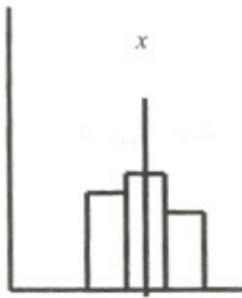
STANDARD: Documenting the use of Mechanical Restraints will include:

- Rationale for use of mechanical restraints.
- Type of restraint used.
- Assessment pre and post application of circulation, sensory, and motor function distal to the restraint every 15 minutes.
- Documentation of report to the receiving hospital.

REFERENCES: North Bay Medical Center Documentation Standard of Care:
Use of Restraints.

Dr. Thomas L. Charron, Health Officer/EMS Agency Medical Director
Effective Date 09-21-00

Dr. Allen Morini, Assistant EMS Agency Medical Director
Distribution Date 09-21-00



Solano County Quality Improvement Committee

PARAMETER: Documentation on the MICN form.

STANDARD: The following are minimum standards when documenting on the MICN form:

- Legibility.
- Date.
- Time.
- Unit #, Provider Agency.
- MICN authorization number.
- Narrative description of event.
- Base assessment.
- Signature of MICN, physician.
- Destination.
- Trauma score (if applicable).
- ER diagnosis and disposition if the Base is also the receiving facility.

Thomas Charron, MD

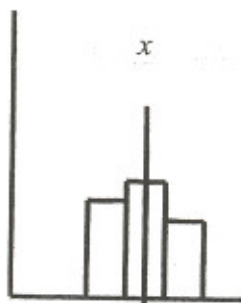
Dr. Thomas L. Charron, Health Officer/EMS Agency Medical Director

Effective Date 09-21-00

A. Morini, MD

Dr. Allen Morini, Assistant EMS Agency Medical Director

Distribution Date 09-21-00



Solano County Quality Improvement Committee

PARAMETER: Respiratory Distress

STANDARD: All patients who present with the complaint of Respiratory Distress will have the following minimum documentation:

- Description of dyspnea to include:
 - a. Time of onset.
 - b. Type of onset.
 - c. Associated cough
 - d. Associated pain.
- Treatment prior to arrival.
- Vital signs to include respiratory pattern.
- Past medical history.
- Primary Medical Doctor.
- History of illegal drug use related to chief complaint.
- Medications noted.
- Allergies noted.
- Treatments given.
- Response to treatment given.

REFERENCE: Bryan E. Bledsoe, Robert S. Porter, Bruce R. Shade, *Paramedic Emergency Care* 1991 Prentice-Hall.

Thomas Charron MD

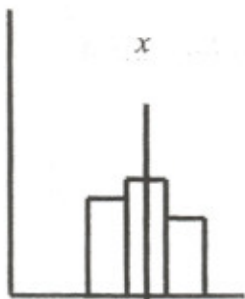
Dr. Thomas L. Charron, Health Officer/EMS Agency Medical Director

Effective Date 09-21-00

A. Morini, MD

Dr. Allen Morini, Assistant EMS Agency Medical Director

Distribution Date 09-21-00



Solano County Quality Improvement Committee

PARAMETER: Altered Level of Consciousness

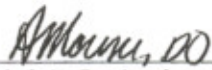
STANDARD: All patients who present with altered level of consciousness will have the following minimum documentation:

- Description of orientation of the patient to include:
 - Level of alertness.
 - Response to stimuli.
- Glasgow Coma Scale.
- Vital signs.
- Description of environment surrounding the patient to include traumatic causes.
- Baseline medical history.
- Past medical history.
- Primary Medical Doctor.
- History of ETOH use related to chief complaint.
- History of drug use related to chief complaint.
- Medications noted.
- Allergies noted.
- Selected and followed appropriate protocol.

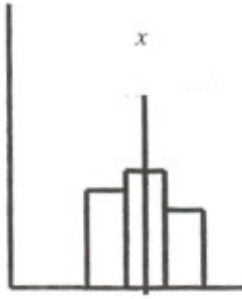
REFERENCE: Bryan E. Bledsoe, Robert S. Porter, Bruce R. Shade, *Paramedic Emergency Care* 1991 Prentice-Hall.



Thomas L. Charron, MD, MPH, Health Officer, EMS Agency Medical Director
Effective Date 09-21-00



Allen Morini, DO, Assistant EMS Agency Medical Director
Distribution Date 09-21-00



Solano County Quality Improvement Committee

PARAMETER: Cervical Spine Immobilization

STANDARD: In general, all patients with blunt trauma, head trauma or axial spine trauma who meet trauma triage criteria require Cervical Spine (C-Spine) precautions.

C-Spine precautions may be **omitted** when **all** of the following conditions apply:

- Normal neurological exam
 - a. alert
 - b. fully oriented to person, place, time and situation.
 - c. normal sensory and motor function in all extremities.
- Absence of neck and/or spinal pain by patient report.
- Absence of neck and/or spinal tenderness elicited by palpation.
- No evidence of impairment by a drug or ETOH.
- Normal vital signs.
- Patient's age is greater than 14 years.
- No history of loss of consciousness.
- Absence of any painful injury that could distract the patient's ability to appreciate pain.

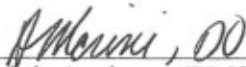
C-Spine stabilization when applied must include:

- Rigid spine board, or similar transporting device.
- Semi-rigid cervical collar.
- Lateral neck rolls or approved stabilization device such as the Headbed ®.
- Tape across the forehead and collar or equivalent.
- Straps across the patient's chest, abdomen, and legs to secure patient to device and prevent movement in any direction.

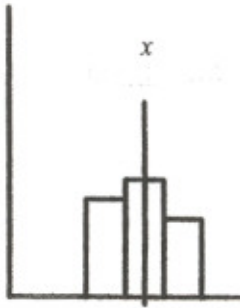
REFERENCES: SPINE INJURY Clinical Criteria for Assessment and Management.
Peter Goth, M.D. Revised May 1995



Dr. Thomas L. Charron, Health Officer/EMS Agency Medical Director
Effective Date 09-21-00



Dr. Allen Morini, Assistant EMS Agency Medical Director
Distribution Date 09-21-00



Solano County Quality Improvement Committee

PARAMETER: Airway/Respiratory Assessment and Treatment

STANDARD: All patients with a respiratory complaint or a *potential* airway problem (i.e., altered LOC, EtOH, burns) will at a minimum have the following assessments and/or treatments documented:

- Adequate airway patency and breathing
- Rate, quality and character of breath sounds
- Appropriate positioning of patient
- Use of proper BLS/ALS airway adjuncts
- Administration of oxygen in a timely fashion, if appropriate, using correct delivery device and flow rates
- Proper ventilation technique/equipment
- Suctioning, if needed
- Frequent reassessment/re-evaluation of the patient and interventions

REFERENCES: Solano County CQI

Thomas Charron M.D.

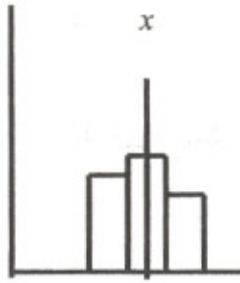
Dr. Thomas L. Charron, Health Officer/EMS Agency Medical Director

10/30/97
Effective Date

Allen J. Morini, D.O.

Dr. Allen Morini, Assistant EMS Agency Medical Director

10/30/97
Distribution Date



Solano County Quality Improvement Committee

PARAMETER: Verification of Out-of-Hospital Endotracheal Tube Placement

- STANDARD:**
1. Mandatory requirements for "Check and Chart":
 - a. **Direct visualization of the chords**
 - b. Chest movement
 - c. Breath sounds
 - d. Epigastric auscultation and observation
 - e. Type of securing device
 - f. Confirmatory maneuvers/devices (see #3 and 4 below)
 - g. **Reassess** after initial intubation and **any time** patient is **moved** (i.e., floor to board, down stairs, into an ambulance, etc.)
 2. Optional evaluation and charting items:
 - a. End-tidal CO₂ measurement
 - b. Tube condensation
 - c. Reservoir bag compliance
 3. All patients with a perfusing rhythm that are endotracheally intubated will have the following confirmatory maneuvers checked and charted:
 - a. Pulse oximetry, if available
 - b. End-tidal CO₂ measurement
 - c. Esophageal detector device (EDD)
 4. All patients in cardiac arrest that are endotracheally intubated will have the following confirmatory maneuvers checked and charted:
 - a. Esophageal detector device (EDD)
 - b. End-tidal CO₂ measurement may be used additionally, but may be unreliable

Note: Success is greatly affected by tube size; smaller tubes (6.5 - 7.0) are preferable.

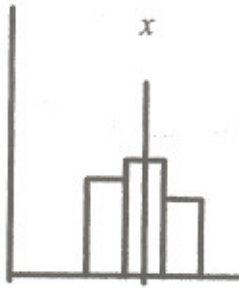
REFERENCE: Position Paper Forum, NAEMSP Standards & Clinical Practice Committee
July 1997 (publication pending)



Dr. Thomas L. Charron, Health Officer/EMS Agency Medical Director
Effective Date 09-21-00



Dr. Allen Morini, Assistant EMS Agency Medical Director
Distribution Date 09-21-00



Solano County Quality Improvement Committee

Parameter: Pulse Oximetry

Standard: When available, the prehospital personnel should use the Pulse Oximeter device to assess any patient with a Respiratory / Cardiac / ALOC / or Trauma as a Chief complaint. The results, intervention and trending should be documented on the PCR. **Oxygen should not be withheld as a treatment based on the pulse oximetry reading.**

Reference: Bledsoe, B.; Porter, R.; Shade, B., *Paramedic Emergency Care*, 1991, Prentice-Hall.

Thomas L. Charron 3-21-02

Dr. Thomas L. Charron, Health Officer/EMS Agency Medical Director
Effective Date 03-21-02

REVIEWED BY PHYSICIANS FORUM MARCH 14, 2002